

JEE MAIN REVISION MASTER: VECTOR ALGEBRA

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 282 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Dot Product, Projections & Orthogonality	116 PYQs	Scalar Product
Very High	Scalar Triple Product (Box Product) & Coplanarity	62 PYQs	Volume & Coplanarity
Very High	Vector Triple Product & Lagrange Identity	39 PYQs	Triple Products
Very High	Cross Product & Geometric Areas	38 PYQs	Vector Product
Very High	Section Formula, Collinearity & Centroids	26 PYQs	Geometry & Position Vectors
Very High	Linear Combination & Vector Equations	20 PYQs	Vector Equations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Dot Product, Projections & Orthogonality	$\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos\theta$, $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$, $\text{Proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{ \vec{b} ^2} \vec{b}$
Scalar Triple Product (Box Product) & Coplanarity	$[\vec{a}, \vec{b}, \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$, $[\vec{a}, \vec{b}, \vec{c}] = 0 \iff \text{coplanar}$
Vector Triple Product & Lagrange Identity	$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$, $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
Cross Product & Geometric Areas	$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$, $\text{Area}_{\triangle} = \frac{1}{2} \vec{a} \times \vec{b} $, $\hat{n} = \frac{\vec{a} \times \vec{b}}{ \vec{a} \times \vec{b} }$
Section Formula, Collinearity & Centroids	$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$, $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$, $\vec{a} = \lambda \vec{b} \iff \text{collinear}$
Linear Combination & Vector Equations	$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$, $\text{or take } \vec{a} \times (\vec{r} \times \vec{b}) \text{ to decouple}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Dot Product, Projections & Orthogonality (Very High Yield — 116 PYQs)

Core Principle: Scalar product $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_x b_x + a_y b_y + a_z b_z$. Orthogonality condition: $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$. Scalar projection of $\vec{a}$ on $\vec{b}$: $\text{Proj}_{\vec{b}}(\vec{a}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$.

■ **Caution / JEE Pitfall:** 1. Vector projection of $\vec{a}$ along $\vec{b}$ is $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\right)\vec{b}$. 2. $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2(\vec{a} \cdot \vec{b})$ (do not write $2|\vec{a}||\vec{b}|$ unless collinear!).

■ **Speed Trick:** Unit vector along the angle bisector of $\vec{a}$ and $\vec{b}$: $\vec{u} = \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{u}$ and $\vec{v}$ be unit vectors inclined at an acute angle such that $|\vec{u} \times \vec{v}| = \frac{\sqrt{3}}{2}$. If $\vec{A} = \dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = 2\vec{i} + \vec{j} - 2\vec{k}$, $\vec{b} = \vec{i} + \vec{j}$ and $\vec{c} = \vec{a} \times \vec{b}$. Let $\vec{d}$ be a vector such that $\dots$ " [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "For three unit vectors $\vec{a}, \vec{b}, \vec{c}$ satisfying $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 = 9$ and $\dots$ " [Ans: N/A]

3.2 Scalar Triple Product (Box Product) & Coplanarity (Very High Yield — 62 PYQs)

Core Principle: Scalar triple product $[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \det([a_i])$. Three vectors are coplanar if and only if $[\vec{a} \vec{b} \vec{c}] = 0$. Volume of parallelepiped $V = |[\vec{a} \vec{b} \vec{c}]|$; tetrahedron volume $V_{\text{tet}} = (1/6)|[\vec{a} \vec{b} \vec{c}]|$.

■ **Caution / JEE Pitfall:** 1. Cyclic order preserves sign: $[\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]$. Non-cyclic swap negates: $[\vec{a} \vec{c} \vec{b}] = -[\vec{a} \vec{b} \vec{c}]$. 2. If any two vectors are identical or collinear, the box product is strictly 0.

■ **Speed Trick:** Four position points A, B, C, D are coplanar if and only if $[\vec{AB} \vec{AC} \vec{AD}] = 0$.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = -\vec{i} + \vec{j} + 2\vec{k}$, $\vec{b} = \vec{i} - \vec{j} - 3\vec{k}$, $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d} = \vec{c} \times \vec{a}$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = -\vec{i} + 2\vec{j} + 2\vec{k}$, $\vec{b} = 8\vec{i} + 7\vec{j} - 3\vec{k}$ and $\vec{c}$ be a vector such that $|\vec{a}| \times |\vec{c}| = |\vec{b}|$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (Numerical): "If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = \vec{j} - \vec{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\dots$ " [Ans: 3]

3.3 Vector Triple Product & Lagrange Identity (Very High Yield — 39 PYQs)

Core Principle: Vector triple product expands via the BAC-CAB identity: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$. It lies in the plane of $\vec{b}$ and $\vec{c}$, perpendicular to $\vec{a}$.

■■ **Caution / JEE Pitfall:** Vector triple product is non-associative: $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$. Notice that $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$.

■ **Speed Trick:** Mnemonic: 'BAC minus CAB' — dot product of outer vectors multiplied by inner vector, minus dot product of first two multiplied by the third.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "Let $\vec{a} = i + 2j + 3k$, $\vec{b} = 3i + j - k$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with..." [Ans: N/A]
- [2025 Evening Shift] Q#4 (MCQ): "Let $\vec{a} = i + 2j + k$ and $\vec{b} = 2i + j - k$. Let $\vec{c}$ be a unit vector in the plane of the vectors $\vec{a}$ and $\vec{b}$ such that $\vec{a} \times \vec{b} = \lambda \vec{c}$." [Ans: N/A]
- [2025 Morning Shift] Q#4 (MCQ): "Let $\vec{a} = 2i - j + 3k$, $\vec{b} = 3i - 5j + k$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{b} = \lambda \vec{c}$." [Ans: N/A]

3.4 Cross Product & Geometric Areas (Very High Yield — 38 PYQs)

Core Principle: Vector cross product $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta \hat{n}$. Collinearity condition: $\vec{a} \parallel \vec{b}$ iff $\vec{a} \times \vec{b} = \vec{0}$. Area of triangle with sides $\vec{a}$, $\vec{b}$ is $(1/2)|\vec{a} \times \vec{b}|$; parallelogram area is $|\vec{a} \times \vec{b}|$; parallelogram with diagonals $\vec{d}_1$, $\vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$.

■■ **Caution / JEE Pitfall:** 1. Cross product is anti-commutative: $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$. 2. Lagrange's identity: $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.

■ **Speed Trick:** Area of a parallelogram given diagonal vectors $\vec{d}_1$, $\vec{d}_2$ is $(1/2)|\vec{d}_1 \times \vec{d}_2|$, NOT $|\vec{d}_1 \times \vec{d}_2|$!

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let P be a point in the plane of the vectors $\vec{AB} = 3i + j - k$ and $\vec{AC} = i - j + 3k$ such that P is equidistant from A , B and C ." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let $\vec{a} = 2i + 3j + 3k$ and $\vec{b} = 6i + 3j + 3k$. Then the square of the area of the triangle with adjacent sides $\vec{a}$ and $\vec{b}$ is..." [Ans: C]
- [2026 Morning Shift] Q#3 (Numerical): "Let PQR be a triangle such that $\vec{PQ} = -2i - j + 2k$ and $\vec{PR} = ai + bj - 4k$, $a, b \in \mathbb{Z}$. Let S be the area of the triangle PQR ." [Ans: 37]

3.5 Section Formula, Collinearity & Centroids (Very High Yield — 26 PYQs)

Core Principle: Section formula for point dividing line segment in ratio $m:n$: $\vec{r} = \frac{n\vec{a} + m\vec{b}}{m+n}$ (internal) and $\vec{r} = \frac{n\vec{a} - m\vec{b}}{m-n}$ (external). Centroid of triangle $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$. Three points are collinear if $\vec{AB} = \lambda \vec{BC}$.

■ **Caution / JEE Pitfall:** Collinear vectors can be antiparallel (negative scalar λ). External division formula carries an explicit minus sign in numerator and denominator: $\frac{n\vec{a} - m\vec{b}}{m-n}$.

■ **Speed Trick:** In a tetrahedron with vertices $\vec{a}, \vec{b}, \vec{c}, \vec{d}$, the centroid is $\vec{G} = \frac{\vec{a} + \vec{b} + \vec{c} + \vec{d}}{4}$.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "Let the lines $L_1: \vec{r} = i\hat{i} + 2j\hat{j} + 3k\hat{k} + \lambda(2i\hat{i} + 3j\hat{j} + 4k\hat{k})$, $\lambda \in \mathbb{R}$ and $L_2: \vec{r} = (4\hat{i} + \hat{j} + \hat{k}) + \mu(2i\hat{i} + 3j\hat{j} + 4k\hat{k})$, $\mu \in \mathbb{R}$ be..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (Numerical): "If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4i\hat{i} - k\hat{k} + \mu(2i\hat{i} + 3k\hat{k})$, $\mu \in \mathbb{R}$ is..." [Ans: 170]
- [2026 Evening Shift] Q#3 (MCQ): "Let O be the origin, $\vec{OP} = \vec{a}$ and $\vec{OQ} = \vec{b}$. If R is the point on $\vec{OP}$ such that $\vec{OP} = 5\vec{OR}$, and $\vec{OR} = \lambda\vec{a} + \mu\vec{b}$..." [Ans: N/A]

3.6 Linear Combination & Vector Equations (Very High Yield — 20 PYQs)

Core Principle: Solving vector equations like $\vec{r} \times \vec{a} = \vec{b}$ and $\vec{r} \cdot \vec{c} = d$. Express $\vec{r}$ as a linear basis expansion $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$ or take vector cross/dot products with known vectors on both sides.

■ **Caution / JEE Pitfall:** When $\vec{r} \times \vec{a} = \vec{b}$, note that $\vec{b} \perp \vec{a}$ (so $\vec{a} \cdot \vec{b} = 0$). The general solution is $\vec{r} = \lambda \vec{a} + \frac{\vec{a} \times \vec{b}}{|\vec{a}|^2}$.

■ **Speed Trick:** If $\vec{r} \times \vec{a} = \vec{b}$, taking the cross product with $\vec{a}$ gives $\vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$, decoupling $\vec{r}$ directly.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{a} = 4i\hat{i} - j\hat{j} + 3k\hat{k}$, $\vec{b} = 10i\hat{i} + 2j\hat{j} - k\hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3\vec{c} = \vec{0}$..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let $\vec{u}$ and $\vec{v}$ be unit vectors inclined at an acute angle such that $|\vec{u} \times \vec{v}| = \frac{\sqrt{3}}{2}$. If $\vec{a} = \vec{u} + \vec{v}$..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Let the vectors $\vec{a} = -i\hat{i} + j\hat{j} + 3k\hat{k}$ and $\vec{b} = i\hat{i} + 3j\hat{j} + k\hat{k}$. For some $\lambda, \mu \in \mathbb{R}$, let $\vec{c} = \lambda\vec{a} + \mu\vec{b}$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

If $\vec{a} = i\hat{i} + j\hat{j} + k\hat{k}$, $\vec{b} = j\hat{j} - k\hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{c} \cdot (\vec{a} - 2\vec{b})$ is equal to ____.

Correct Answer: 3

Solution Analysis: Let $\vec{a} = i\hat{i} + j\hat{j} + k\hat{k}$, $\vec{b} = j\hat{j} - k\hat{k}$. We have to find $\vec{c} \cdot (\vec{a} - 2\vec{b})$ given that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$. Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$. Then $\vec{a} \times \vec{c} = (1, 1, 1) \times (x, y, z) = (z - y, x - z, y - x)$. Now compute $\vec{a} \times \vec{c} = \vec{b} = (0, 1, -1)$. So, $(z - y, x - z, y - x) = (0, 1, -1)$. Expanding, $\vec{a} \times \vec{c} = i(z - y) - j(z - x) + k(y - x)$. Since this is equal to $\vec{b} = j - k$, we compare components: $z - y = 0$, $x - z = 1$, $y - x = -1$. $\Rightarrow$

Benchmark Problem 2: JEE Main 2026 (Online) 6th April Evening Shift (Q#2)

Let $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$ and $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $(2\vec{a} + 3\vec{b})$ and $(\vec{a} - \vec{b})$ is :

- (A) 450
- (B) 900
- (C) 1800 [Correct]
- (D) 2400

Correct Answer: C

Solution Analysis: Given $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$, $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$. We need the square of the area of the triangle whose adjacent sides are $2\vec{a} + 3\vec{b}$ and $\vec{a} - \vec{b}$. For two adjacent sides $\vec{u}$ and $\vec{v}$, area of triangle is $(1/2)|\vec{u} \times \vec{v}|$. So, square of the area is $((1/2)|\vec{u} \times \vec{v}|)^2 = (1/4)|\vec{u} \times \vec{v}|^2$. Let $\vec{u} = 2\vec{a} + 3\vec{b}$, $\vec{v} = \vec{a} - \vec{b}$. First find $\vec{u}$: $2\vec{a} = 2(2\hat{i} + 3\hat{j} + 3\hat{k}) = 4\hat{i} + 6\hat{j} + 6\hat{k}$
 $3\vec{b} = 3(6\hat{i} + 3\hat{j} + 3\hat{k}) = 18\hat{i} + 9\hat{j} + 9\hat{k}$

Benchmark Problem 3: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - k\hat{j} + \mu(2\hat{i} + 3\hat{k})$, $\mu \in \mathbb{R}$ along a line with direction ratios 3, -1, 0 is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Correct Answer: 170

Solution Analysis: Point P : $(43, \alpha, \beta)$ with the constraint $\beta < 0$. Target Line (L_1) : Given by $\vec{r} = 4\hat{i} - k\hat{j} + \mu(2\hat{i} + 3\hat{k})$. A general point Q on this line is $(4 + 2\mu, 0, -1 + 3\mu)$. Direction of measurement : The distance is measured along a line with direction ratios (3, -1, 0). Find Point Q on the Target Line Since the distance is measured along the direction $d = (3, -1, 0)$, the vector PQ must be parallel to d. $PQ = Q - P = (4 + 2\mu - 43, 0 - \alpha, -1 + 3\mu - \beta)$ $PQ = (2\mu - 39, -\alpha, 3\mu - \beta - 1)$ $PQ = (2\mu - 39, -\alpha, 3\mu - \beta - 1)$

Benchmark Problem 4: JEE Main 2025 (Online) 22nd January Morning Shift (Q#1)

Let $\vec{c}$ be the projection vector of $\vec{b} = \lambda\hat{i} + 4\hat{k}$, $\lambda > 0$, on the vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$. If $|\vec{a} + \vec{c}| = 7$, then the area of the parallelogram formed by the vectors $\vec{b}$ and $\vec{c}$ is _____.

Correct Answer: 16

Solution Analysis: To find the projection vector $\vec{c}$ of $\vec{b} = \lambda\hat{i} + 4\hat{k}$ (where $\lambda > 0$) onto vector $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$, we use the formula for the projection of a vector: $\vec{c} = \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$. Calculate the dot product $\vec{b} \cdot \vec{a}$: $\vec{b} \cdot \vec{a} = (\lambda\hat{i} + 4\hat{k}) \cdot (\hat{i} + 2\hat{j} + 2\hat{k}) = \lambda \cdot 1 + 4 \cdot 2 = \lambda + 8$. Calculate the magnitude of $\vec{a}$: $|\vec{a}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$. Now substitute to find $\vec{c}$:
 $\vec{c} = \frac{(\lambda + 8)}{9}(\hat{i} + 2\hat{j} + 2\hat{k})$

Benchmark Problem 5: JEE Main 2024 (Online) 8th April Morning Shift (Q#2)

Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{593|\vec{r}|^2}{593}$ is equal to _____.

Correct Answer: 569

Solution Analysis: $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$ $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$
 $(\vec{r} - (\vec{b} + \vec{c})) \times \vec{a} = 0 \Rightarrow \vec{r} = (\vec{b} + \vec{c}) + \lambda \vec{a}$
 $\vec{r} = (20\hat{i} + 5\hat{j} - 12\hat{k}) + \lambda(9\hat{i} - 13\hat{j} + 25\hat{k}) = (20 + 9\lambda)\hat{i} + (5 - 13\lambda)\hat{j} + (25\lambda - 12)\hat{k}$ Now $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$
 $(-14\hat{i} + 9\hat{j} - 14\hat{k}) \cdot \vec{r} = 0$ Now $\begin{aligned} & -14(20 + 9\lambda) + 9(5 - 13\lambda) - 14(25\lambda - 12) = 0 \\ & -280 - 126\lambda + 45 - 117\lambda - 350\lambda + 168 = 0 \\ & -687\lambda - 167 = 0 \end{aligned}$